



CYBER SECURITY ANALYST INTERNSHIP ENGAGEMENT REPORT

Week 02: Computer Networking Fundamentals & Network Configuration

ORGANIZATION	IT SIMPLERA
REPORT TITLE	Week 02: Computer Networking Fundamentals & Network Configuration
STUDENT NAME	Dua Najam
INSTRUCTOR	Sir Talal Rehan
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1. Executive Summary

This report covers the fundamentals of computer networking through hands-on practice with essential commands. We used **ipconfig /all** to view network settings, **ping** to test connectivity, **tracert** to trace data routes, **nslookup** to find IP addresses, and **arp -a** to list local devices. These tools help identify and fix network problems quickly.

The report also compares **routers and switches**, explaining that routers connect different networks while switches connect devices within the same network. Additionally, it highlights the differences between **IPv4 and IPv6** addressing, and describes the **seven layers of the OSI model** with their functions, protocols, and devices, giving a clear picture of how network communication works.

Task Completion Status

Task	Description	Status
Task 01	Network Information	Completed
Task 02	Basic Networking Commands	Completed
Task 03	Router VS Switch	Completed
Task 04	IPV4 VS IPV6	Completed
Task 05	OSI Model	Completed

2. Learning Objectives

1. **Understand basic networking concepts** including IP addresses, MAC addresses, and default gateways.
2. **Learn to use `ipconfig /all`** to identify and interpret computer network configuration.
3. **Execute essential networking commands** such as ping, tracert, nslookup, and arp to diagnose network issues.
4. **Differentiate between routers and switches** by comparing their functions, OSI layers, and use cases.
5. **Compare IPv4 and IPv6** addressing schemes and identify valid examples of each.
6. **Explain the seven layers of the OSI Model** with their functions, protocols, and associated devices.
7. **Develop practical troubleshooting skills** through hands-on command execution and technical documentation.

3. Task Documentation

3.1 Task 01 — Network Information

- **ipconfig /all:** It shows full network details of your computer, including IP addresses, MAC addresses, and adapter info.
 - ➔ **IPS are hidden due to privacy and security**

```
C:\Users\cisco>ipconfig /all

Windows IP Configuration

Host Name . . . . . : cisco-pc21b
Primary Dns Suffix . . . . . : ssuet.edu.pk
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : ssuet.edu.pk

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : ssuet.edu.pk
Description . . . . . : Realtek PCIe GbE Family Controller
Physical Address. . . . . : [REDACTED]
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::6491:6823:ad79:42db%11(Preferred)
IPv4 Address. . . . . : [REDACTED](Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, 28 October 2025 11:36:16 am
Lease Expires . . . . . : Tuesday, 28 October 2025 7:36:16 pm
Default Gateway . . . . . : [REDACTED]
DHCP Server . . . . . : [REDACTED]
DHCPv6 IAID . . . . . : 214489554
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-BC-59-A6-C8-D9-D2-35-34-ED
DNS Servers . . . . . : [REDACTED]

NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 58-A0-23-67-F0-49
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 5A-A0-23-67-F0-48
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

FIELD	Purpose
IPv4 Address	Your computer's unique address on the network. Other devices use this to send data to your computer.
MAC Address	Your computer's permanent hardware ID. The network uses this to identify your device locally.
Default Gateway	It is the router's address. Your computer sends all internet traffic through this gateway.

EXTRA INFORMATION

FIELD	PURPOSE
Host Name	Your computer's name
DNS Suffix	Domain Name
Subnet Mask	Network range
DHCP Server	Server that gave you IP
DNS Servers	Servers that translate website names to IP

3.2 Task 02 — Basic Network Commands

1. **ipconfig /all**: It shows full network details of your computer, including IP addresses, MAC addresses, and adapter info.
2. **ping google.com**: The ping command checks if your computer can connect to another device or website and shows how fast the response is.

```
C:\Users\cisco>ping www.google.com

Pinging www.google.com [216.239.38.120] with 32 bytes of data:
Reply from 216.239.38.120: bytes=32 time=14ms TTL=115
Reply from 216.239.38.120: bytes=32 time=14ms TTL=115
Reply from 216.239.38.120: bytes=32 time=14ms TTL=115
Reply from 216.239.38.120: bytes=32 time=14ms TTL=115

Ping statistics for 216.239.38.120:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 14ms, Average = 14ms
```

3. **tracert google.com**: The tracert (or traceroute) command shows the path your data takes to reach a website or IP address, listing all the routers it passes through.

```
C:\Users\cisco>tracert www.google.com

Tracing route to www.google.com [216.239.38.120]
over a maximum of 30 hops:

  0  2 ms    2 ms    2 ms  10.2.14.1
  1  <1 ms   <1 ms   <1 ms  192.168.2.1
  2  <1 ms   11 ms   <1 ms  203.215.185.38
  3  1 ms    1 ms    1 ms  192.168.11.61
  4  1 ms    1 ms    1 ms  119.63.137.181
  5  2 ms    3 ms    2 ms  110.93.252.172
  6  2 ms    8 ms    8 ms  110.93.252.246
  7  15 ms   15 ms   15 ms  110.93.192.35
  8  15 ms   15 ms   15 ms  209.85.241.201
  9  15 ms   15 ms   15 ms  142.251.50.209
 10  15 ms   15 ms   14 ms  any-in-2678.1e100.net [216.239.38.120]

Trace complete.
```

4. **nslookup google.com**: This command lets you find the IP address of a website or check the domain name associated with an IP address.

```
C:\Users\cisco>nslookup www.google.com
Server: JINNAH.ssuet.edu.pk
Address:

Name: www.google.com
Address: 216.239.38.120
```

5. **arp -a**: Displays all current ARP entries for all network interfaces.

```
C:\Users\cisco>arp -a

Interface: 10.2.14.33 --- 0xb

Internet Address      Physical Address      Type
10.2.14.1             08-cc-a7-29-52-d6     dynamic
10.2.14.255           ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

3.3 Task 03 — Router VS Switch

A router connects different networks and uses IP addresses at Layer 3. A switch connects devices within the same network and uses MAC addresses at Layer 2. Routers are expensive and slower, while switches are cheaper and faster.

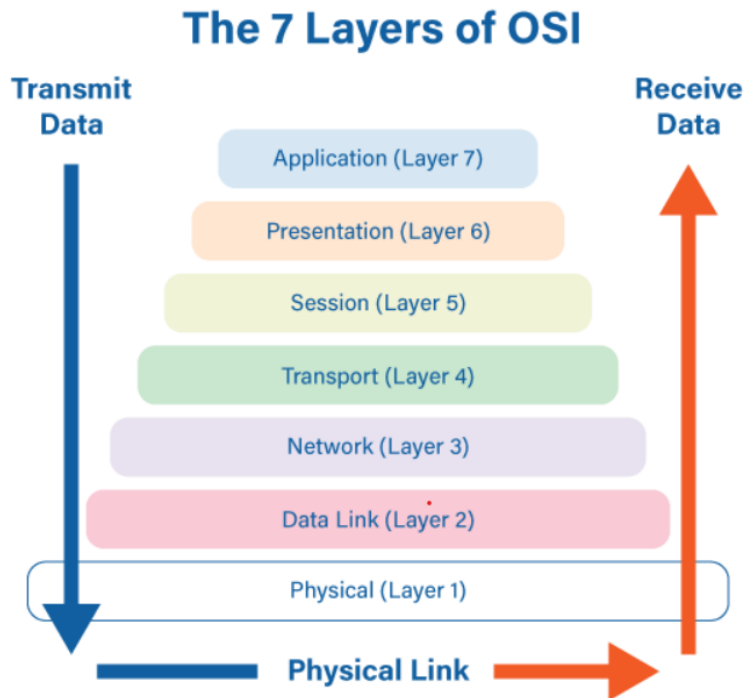
COMPARISION POINT	ROUTER	SWITCH
Function	Connects different networks together	Connects devices within the same network
OSI Layer	Layer 3 (Network Layer)	Layer 2 (Data Link Layer)
Addressing	Uses IP address	Uses MAC address
Use Case	Sharing internet, connecting remote networks	Connecting computers, printers in an office
Speed	Slower	Faster
Price	Expensive	Cheaper

3.4 Task 04 — IPV4 VS IPV6

IPv4 uses 32-bit addresses with 4 numbers separated by dots, while IPv6 uses 128-bit addresses with 8 groups of hexadecimal characters separated by colons. IPv4 can support about 4.3 billion devices, but IPv6 supports a much larger number of devices. IPv4 is still widely used today, while IPv6 is newer and designed for future internet growth.

COMPARISION POINT	IPv4	IPv6
Address Size	32-bit	128-bit
Format	4 numbers separated by dots	8 groups separated by colons
Example	192.168.1.1	2001:0db8:85a3:0000:0000:8a2e:0370:7334
Number of Devices	About 4.3 billion	Almost unlimited
Security	Less secure	More secure with built-in features
Usage	Older, widely used today	Newer, designed for future

3.5 Task 05 — OSI Model



LAYER	NAME	FUNCTION	PROTOCOL	DEVICE
7	Application	Shows data to user, runs apps	HTTP, FTP	Computer, Phone
6	Presentation	Converts data to readable format	SSL, JPEG	Gateway
5	Session	Starts, manages, and ends sessions	NetBIOS, RPC	Gateway
4	Transport	Breaks data into packets, ensures delivery	TCP, UDP	Firewall

LAYER	NAME	FUNCTION	PROTOCOL	DEVICE
3	Network	Finds best route to send data	IP, ICMP	Router
2	Data Link	Direct connection between devices	Ethernet, PPP	Switch
1	Physical	Handles wires, signals, and hardware	USB, Bluetooth	Cable, Hub

4. Challenges Faced

1. **Understanding command outputs** was initially difficult due to technical terminology and multiple parameters displayed.
2. **Differentiating between routers and switches** required careful study as both devices appear similar but function differently.
3. **Memorizing the seven OSI layers** in correct order was challenging without proper mnemonics.
4. **Interpreting IPv6 addresses** was complicated due to their longer and more complex format compared to IPv4.

5. Conclusion

This lab provided valuable hands-on experience with computer networking concepts. By practicing essential commands and studying network devices, addressing schemes, and the OSI model, we gained practical skills for troubleshooting and understanding network communication. These fundamentals are crucial for managing and maintaining modern computer networks.

6. References

1. Tanenbaum, A. S., & Wetherall, D. J. (2011). *Computer Networks* (5th ed.). Pearson.
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